Contrastive attack for deepfake protection

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Abstract

While the remarkable advancement of generative AI has brought significant benefits, it has also given rise to misuse, particularly in the form of crimes utilizing generative models. Specifically, face swapping, known as 'deepfakes,' has become a serious threat not only to celebrities but also to ordinary individuals. Existing countermeasures primarily focus on passive detection, which is often ineffective for media not yet distributed on public platforms. Although some active defense methods have been proposed, most rely on white-box assumptions or GAN-based models. These approaches frequently suffer from training instability and generate low-fidelity images, leading to limited reproducibility. To address these challenges, we propose a novel contrastive attack method that adds adversarial noise to images under a black-box assumption. Unlike methods that attack the entire face-swapping pipeline, our approach targets the face recognition models used as identity encoders, making it highly adaptable to black-box scenarios. Furthermore, we employ a contrastive learning loss to maximize the distance between original and perturbed images, eliminating the need for specific target images. Across three black-box face swappers, our training-free attack delivers competitive protection with a strictly bounded, reproducible perturbation that preserves facial geometry, whereas our faithful re-implementation of a recent generator-based state of the art fails to reproduce its reported fidelity and instead distorts facial structure. We demonstrate the effectiveness of our method through quantitative metrics and qualitative visualizations.

1. Introduction

Recently, there has been remarkable advancement in artificial intelligence, particularly with deep neural networks. While these advancements have brought significant benefits to generative AI, they also pose serious risks. The misuse of generative models has surged, leading to crimes such as the abuse of language models, the spread of fake news, and deepfake generation. In particular, deepfake generation has become a severe threat not only to celebrities but also to ordinary individuals. Face swapping models [3, 30, 32, 41], commonly known as 'deepfakes,' replace the facial identity of a target image with that of a source victim.

Existing countermeasures against face swapping largely focus on passively detecting manipulated images [13, 15, 29, 48] rather than actively protecting source images. Recently, a few approaches have explored proactive protection using adversarial attack methods [1, 7, 14, 17, 18, 31, 33, 35, 37, 39, 40, 44, 47, 50]. However, these methods suffer from limited transferability due to redundant attacks on full face swapping pipelines [1, 33, 37], visual distortion of the original image with restricted reproducibility in GAN-based models [7, 18, 31, 39, 44], reliance on grey-box scenarios with universal noise patterns [17], or the need to handle complex latent geometries [14, 35].

To deploy prevention methods in the real world, several criteria must be met: i) eliminating the need for a target image, which is often inaccessible or requires reproducing another victim; ii) ensuring that perturbations and visual transformations remain small as possible or even imperceptible to humans; and iii) adhering to a black-box setting effective against unseen face swapping models. If the goal is to alter the latent identity of the source image, we only need to attack the facial identity feature extraction module rather than the entire face swapping model. Since most face swapping architectures utilize face recognition models as identity feature extractors [3, 32, 41], there is no need to attack the full pipeline when the objective is solely to prevent identity extraction.

To tackle these issues and provide a camera-ready prevention method, we propose a novel contrastive attack. We eliminate the need for target images by defining the victim's opposite latent vector using a contrastive loss [4] accompanied by augmentations of the original face feature vector. Furthermore, we stabilize the attack phase and maximize the distance between the original and perturbed images by leveraging the characteristics of latent geometry.

2. Related Works

2.1. Face Swapping

Face swapping aims to transfer the identity of a source face to a target face while preserving the target's attributes, such as expression, pose, and background. Early approaches [24, 30] primarily relied on Auto-Encoder (AE) [23] architectures. While effective, these methods typically require training separate models for specific pairs of identities, limiting their application to arbitrary faces.

To address this limitation, recent research has shifted towards subject-agnostic frameworks capable of one-shot face swapping. SimSwap [3], a work brought significant impact, introduced an ID injection module utilizing Adaptive Instance Normalization (AdaIN)[19] to transfer identity information into the target generation process without retraining. Following this pipeline, Gao *et al*. [11] applied the Information Bottleneck principle to enhance the disentanglement between identity and attribute features. Similarly, UniFace [43] proposed a unified framework that jointly performs face swapping and reenactment, exploiting the intrinsic relationship between the two tasks to improve feature disentanglement.

More recent studies focus on high-fidelity generation and robustness against complex scenarios. HifiFace [41] incorporates 3D shape and semantic priors to maintain structural consistency. To handle challenging conditions such as occlusions and extreme poses, Facedancer [32] introduced an occlusion-aware framework. Furthermore, to achieve fine-grained control, Liu *et al*. [27] utilized regional GAN inversion, while Wang *et al*. [38] proposed a landmark-guided feature entanglement module to effectively preserve non-identity attributes.

2.2. Active Defense against Deepfakes

Unlike passive detection methods that identify forged images post-hoc, active defense mechanisms—also known as proactive perturbation—aim to disrupt the Deepfake generation process by injecting imperceptible adversarial noise into source images.

2.2.1. Gradient-based Early Approaches

Early approaches employed gradient-based optimization to insert perturbations directly into the image pixel space. Ruiz *et al*. [33] proposed a class-transferable adversarial attack to distort the output of conditional image translation networks. Similarly, Aneja *et al*. [1] introduced targeted attacks (TAFIM) to manipulate facial expressions or attributes towards a specific target. To improve visual imperceptibility, Wang *et al*. [37] explored adding perturbations to the luminance channel of the Lab color space.

2.2.2. Generator-based and Robust Approaches

To enhance attack transferability and handle black-box scenarios, recent studies have shifted towards generator-based methods. Huang *et al*. [18] proposed an initiative defense framework by training a surrogate model and a poison perturbation generator. Dong *et al*. [7] utilized a GAN-based architecture to construct perturbations capable of disrupting unknown Deepfake systems. Furthermore, several works focus on robustness against real-world distortions. Qu *et al*. [31] designed DF-RAP with a compression-approximation GAN to withstand social network compressions, while Huang *et al*. [17] introduced a cross-model universal watermark (CMUA) to attack multiple models iteratively. More recently, Wang *et al*. introduced NullSwap[39] which achieves State Of The Art performance by utilizing identity discriminator with Dynamic Loss Weighting regime. However, these generator-based approaches often require complex training procedures (e.g., GAN training), which are prone to instability and reproducibility issues. Moreover, they frequently rely on white-box assumptions or extensive surrogate model training, incurring high computational costs.

2.2.3. Feature-level Disruption

Recognizing that most Deepfake models rely on extracting identity or attribute features, recent state-of-the-art methods target the intermediate feature representations rather than the final output. He *et al*. [14] proposed disrupting Deepfakes by searching for adversarial embeddings in the latent space that lead to failed reconstructions. More recently, Tang *et al*. [35] introduced FOUND, a feature-output ensemble disruptor that attacks feature extraction modules by maximizing the Wasserstein distance between features.

While existing feature-level attacks have shown promise, they often involve complex ensemble strategies or optimization processes. In contrast, our work proposes a simplified yet effective contrastive attack[4] tailored for black-box scenarios. Instead of attacking the entire Deepfake pipeline or training unstable generative models, we specifically target the face recognition model used as the identity encoder in various Deepfake systems. By employing a contrastive learning loss[4], we push the features of the perturbed image away from the original one without requiring any target images or white-box access to the generative network. This approach ensures high transferability and training stability with minimal computational overhead.

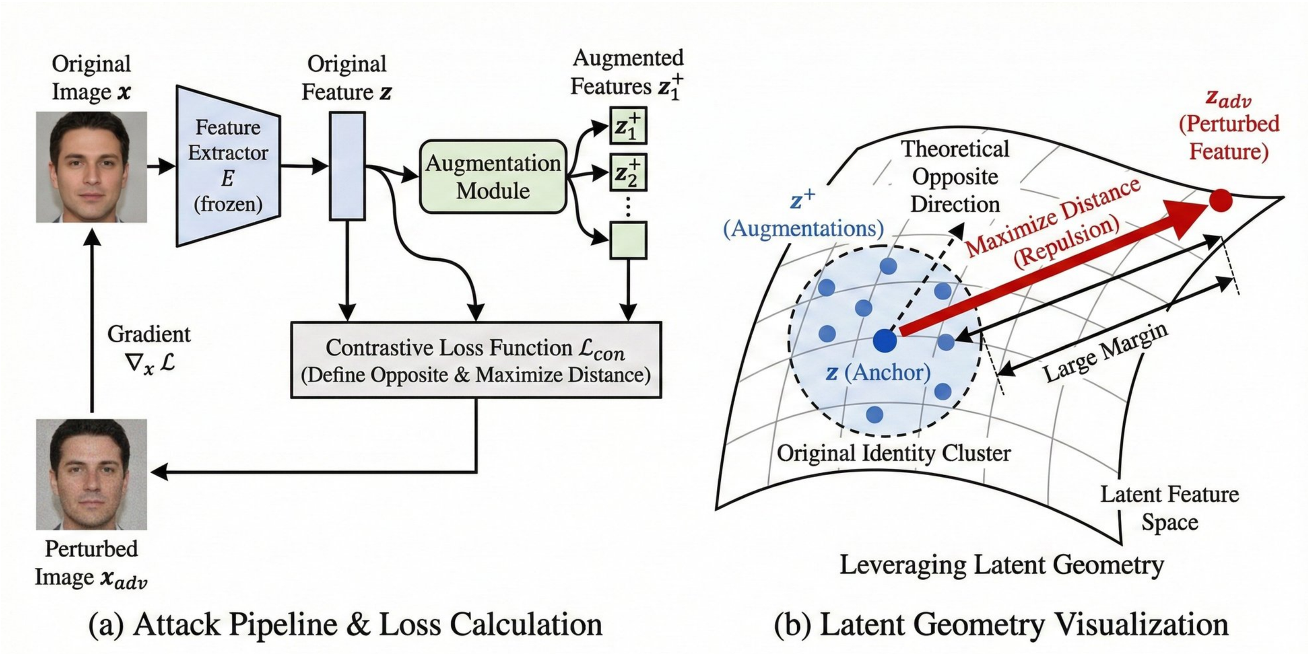

Figure 1. Visualization of our contrastive attack method (a) represents abstract pipeline of our method (b) shows abstract geometry during attack

3. Method

3.1. Problem Formulation

Although the key to victory against face swapping model is interrupting extraction facial identity feature for encoder module, many of prevention methods attend on attacks for full face swapping models instead of only encoder parts. Aforementioned, the modern face swapping models utilize face recognition model such as Arcface[5] as a identity encoder ever since milestone model SimSwap[3]. Recent method, NullSwap[39] focuses only on face recognition module with its GAN-based regenerating image module. But, it is hard to reproduce the quality mentioned in paper due to instability from GAN training regime and often transports visual features which are small difference in pixel space such as adding or removing mustache, resizing nose. Those make hard to bring prevention method in the real world.

In this manner, gradient based methods bring advantages on its stability with consistency of visual features although noise can be perceptible. But, there are serious drawback of naive gradient based methods caused by characteristic of face recognition model which only gives latent vector instead of logits.

i) it needs another victim for attack. Not only the reason it produces another victim but, also many cases users don't have access to victim which latent facial features are far enough from user.

ii) The target locates near orthogonal which is often not enough far for black-box scenario. The random vectors locates near orthogonal while increasing dimension[2]. Because the output of face recognition models are high dimensional vectors i.e. 512, they are also located in orthogonal spaces. In addition to that, approaches before [5] like [34] that doesn't consider making margin in latent space have less angles between victim faces. For those reasons, it often fails to transfer into black-box models used by face swapping models.

To solve those issues with leveraging benefits from gradient-based attack, we propose novel contrastive attack that eliminates needs of victim and uses far enough attack objective by setting opposite vector with contrastive loss from facial attribute augmented images.

3.2. FGSM-baseline

We use well-known and effective gradient-based attack baseline which is Fast Gradient Sign Method(FGSM)[12]. It generates noise based on gradient processed by sign function denoted as

$$x^{adv} = x + \alpha \cdot sign(\nabla_x J(\theta, x, y)) \quad (1)$$

where α is hyperparameter for step size, y is a true label of input x. Practically, most cases use iterative approach called Iterative-FGSM(I-FGSM) also knwo as Projected Gradient Descent[25, 28] gradient descent with least-likelihood class[25] instead of gradient ascent with its true label denoted as

$$x_{t+1}^{adv} = Clip_{x,\epsilon}(x_t^{adv} - \alpha \cdot sign(\nabla_x J(\theta, x, y_{ll}))) \quad (2)$$

where ϵ is clipping factor that limits maximum changes of pixel from original image. Like modern optimization methods like Adam[22], FGSM methods also adapt momentum to avoid local minima and increase transferability which known as Momentum Iterative(MI)-FGSM[8].

$$g_{t+1} = \mu \cdot g_t + \frac{\nabla_x J(\theta, x_t^{adv}, y_{ll)}}{||\nabla_x J(\theta, x_t^{adv}, y_{ll)}||_1} \quad (3)$$

$$x_{t+1}^{adv} = Clip_{x,\epsilon}(x_t^{adv} - \alpha \cdot sign(g_{t+1})) \quad (4)$$

But those methods still struggled with transferability for black-box models. Recent methods tried to solve it with various approaches. The most famous and well-known methods are smoothing the gradient with pre-defined kernel which known as Translation Invariant(TI)-FGSM[9], Adding input diversity by random resizing and zero padding which know as Diverse Input(DI)-FGSM[42], ensemble attack[26] by utilizing various models concurrently and using direct logit of least-likelihood class(or certain target) instead of cross-entropy loss by eliminating softmax of loss to avoid gradient vanishing in deep networks which known as logit attack[49]. Combined equation is denoted as

$$\nabla_{temp} = \sum_{k=1}^{N} \nabla_x J_{logit}(\theta_k, T(x_t^{adv}, p), y_{ll}) \quad (5)$$

$$g_{t+1} = \mu \cdot g_t + \frac{W * \nabla_{temp}}{||W * \nabla_{temp}||_1} \quad (6)$$

Where T is a random resizing and padding with probability p, W is a predefined convolution kernel such as gaussian kernel and N is a number of different models for ensemble attack[1].

In contrast to FGSM baseline, there is no certain class for trained face recognition model. So, we use cosine similarity as our objective function. To certain attack on only facial parts of image and getting correct feature with upright face, we crop and rotate image with facial box and landmarks by utilizing face detection model. To guarantee differentiability of image processed with crop and rotation, we compute its affine transformation matrix and warp with sampling matrix derived by affine transformation matrix. The facial attribute can be removed when adapting DI-FGSM. Thus, we add random noise of affine translation matrix to replace MI-FGSM pipeline. Then the temporal gradient can be computed as

$$\nabla_{temp} = \sum_{k=1}^{N} S(\hat{\Theta}(\xi))^T \nabla_x CS(\theta_k, S(\hat{\Theta}(\xi)) x_t^{adv}, y_{target}) \quad (7)$$

Where $\hat{\Theta}(\xi)$ is a affine transformation matrix with added noise parameterized by ξ, S is sampling matrix derived from affine transformation matrix and transpose of matrix is applied for inverse transform of image.

3.3. Contrastive Attack

As discussed in the previous section, standard gradient-based attacks (e.g., FGSM pipeline) necessitate a specific victim identity to guide the optimization. To eliminate this dependency and achieve a victim-free defense, a straightforward intuition is to set the antipodal point (the mathematically opposite vector) of the source embedding as the attack objective. However, directly optimizing towards the exact opposite vector presents a significant theoretical challenge. As derived in prior studies on cosine similarity optimization [10, 45], the gradient of the cosine loss with respect to the input x is defined as:

$$\nabla_x \cos(x, y) = \frac{1}{||x||}(I - \frac{x \bullet x}{||x||^2}) \bullet \frac{y}{||y||} \quad (8)$$

$$||\nabla_x|| \propto \frac{\sin(\angle(x, y))}{||x||} \quad (9)$$

Crucially, as shown in Eq. 9, the magnitude of the gradient is proportional to the sine of the angle between the current embedding x and the target y. When the target is set to the exact opposite vector ($y = -x$), the angle becomes $180°$ (π radians), resulting in $\sin(\pi) = 0$. Consequently, the gradient vanishes, and the input image remains unchanged despite the optimization iterations. This creates a saddle point where the attack fails to initiate.

To overcome this gradient vanishing problem, we exploit the inherent limitations of face recognition models. State-of-the-art models are not perfectly invariant to facial attribute changes; embeddings of the same identity with different attributes (e.g., expressions, accessories, hair color) do not map to a single point but rather form a cluster or manifold in the latent space. This implies that while the gradient flow is blocked along the direct antipodal axis, it remains active for these attribute-variant neighbors.

Therefore, instead of relying on a single source vector, we can leverage these neighboring embeddings to "push" the optimization trajectory away from the source identity's manifold. While a sequential approach—conducting gradient ascent to escape the local manifold followed by gradient descent towards the opposite vector—is possible, it is computationally inefficient and unstable. Instead, we propose a

[1]There is no need to divide gradient by number of models due to sign attack.

unified **Contrastive Attack** framework that leverages both objectives concurrently. We employ a contrastive learning strategy [4] where the adversarial image acts as the anchor, the antipodal vector serves as the positive key (to attract), and the original image along with a set of attribute-variant views serve as negative keys (to repel). Crucially, rather than synthesizing attribute changes with an auxiliary generative model, we obtain attribute-variant neighbors from data that are already available at protection time: (i) a *same-identity pair image*—a different photograph of the same person, which naturally differs in expression, pose, and illumination—and (ii) lightweight input-diversity augmentations of both the source and the pair image, namely horizontal flipping and random resized cropping (RRC). Because these neighbors lie on the same-identity manifold but at a non-zero angle from the source embedding, they surround the source with negatives that provide non-zero gradients and steer the attack out of the antipodal saddle point. This design removes the dependence on a pre-trained attribute-transfer network entirely and improves stability, since the negatives are exact face-recognition embeddings rather than GAN-synthesized approximations.

We modify the standard InfoNCE loss for our adversarial objective as follows:

$$J = -\mathbb{E}\left[\log \frac{\exp(\mathrm{sim}(\theta(x_t), \theta(x_{Opposite}))/\tau)}{\sum_{x^- \in \mathcal{N}} \exp(\mathrm{sim}(\theta(x_t), \theta(x^-))/\tau)}\right] \quad (10)$$

where $\theta(x_{Opposite}) = -\theta(x_{org})$ is the antipodal (positive) key, τ is the temperature, and the negative set is $\mathcal{N} = \{x_{org}, x_{pair}\} \cup \{T(x_{org}), T(x_{pair})\}$, with x_{pair} the same-identity pair image and $T(\cdot)$ the horizontal-flip / RRC input-diversity transform (the negatives are re-encoded each iteration under `no_grad`). This formulation ensures that the adversarial perturbation is optimized to move towards the antipodal direction while maximally diverging from the source identity's attribute manifold. When more than one pair image is available, the negative set extends naturally to all pairs and their augmented views.

Finally, to keep the perturbation perceptually subtle we add an imperceptibility regularizer $R(\cdot,\cdot)$ between the clean image x and the perturbed image $x+\delta$, giving the total objective

$$\mathcal{L} = J + \lambda_{reg}\, R(x,\, x+\delta). \quad (11)$$

R can be a structural-dissimilarity term $1 - \mathrm{SSIM}(x, x+\delta)$, an LPIPS [46] perceptual distance, or a wavelet low-frequency penalty; λ_{reg} trades off protection strength against imperceptibility. We use SSIM by default and study the choice of R as an ablation.

4. Experiments

4.1. Implementation Details

Datasets and Preprocessing. We conduct our experiments on the high-quality facial dataset, CelebA-HQ [20], drawing roughly 1,000 source identities from the held-out validation range (992 evaluated samples). Since our contrastive objective requires attribute-variant neighbors of the source identity, we pair each source with a different image of the *same* identity, using the canonical CelebA-HQ→CelebA identity annotations for a deterministic pairing. To assess out-of-distribution transfer, we additionally evaluate on LFW [16] in a cross-dataset setting, forming same-identity pairs from its identity labels. All images are processed at 256×256; we employ the RetinaFace [6] detector to extract five-point landmarks, which drive a differentiable affine warp to a canonical 112×112 crop for the recognizers. For optimization, pixel values are normalized to the range $[-1, 1]$.

Backbone Models and Ensemble. To ensure robust transferability across black-box face swappers, we employ an ensemble of diverse face recognition models as our surrogate gradients. Specifically, we utilize: (1) two **ArcFace [5]** iResNet-50 backbones that enforce angular margins in the hypersphere but are trained on different corpora (MS1MV3 and Glint360K), providing complementary gradients; (2) **AdaFace [21]** (iResNet-50, WebFace4M), which adapts its margin to image quality. AdaFace is fed BGR inputs per its convention. To further stabilize the gradient, we incorporate the gradients from mirrored (horizontally flipped) versions of the input images. The attribute-variant negatives required by our Contrastive Attack are obtained *directly from data*, not from a generative model: for each source we use one same-identity pair image together with flip / random-resized-crop views of the source and the pair (Sec. 3.3). FaceNet [34] is deliberately *excluded* from the surrogate ensemble and reserved as a held-out evaluator.

Hyperparameters. Our attack is optimized with the MI-FGSM framework under an ℓ_∞ budget of $\epsilon = 8/255$ and momentum decay $\mu = 1.0$, for $T = 300$ iterations with step size 10^{-3} in the normalized $[-1, 1]$ domain. The InfoNCE temperature τ (10) is 1.0. Our default ("Ours") configuration uses one same-identity pair, RetinaFace five-point differentiable-warp alignment of the optimization crop, and an SSIM regularizer with weight $\lambda_{reg} = 0.1$ (11). We further study several axes as ablations, each toggled independently: the imperceptibility regularizer (SSIM / LPIPS / wavelet), TI gradient smoothing [9] with a 5×5 Gaussian kernel, the number of same-identity pairs (1 vs. 5), and a SupCon multi-positive variant. Evaluation always applies the RetinaFace alignment before each recognizer.

4.2. Evaluation

4.2.1. Experimental Setup and Metrics

To evaluate the effectiveness of our defense in a strictly black-box scenario, we employ **QmagFace** [36] as the evaluator model (with FaceNet as a secondary held-out evaluator), ensuring it overlaps with neither the surrogate models used for attack generation nor the internal encoders of the face swappers. We report performance using four families of metrics:

i) **Identity Similarity (id-sim) and NCS**: Our primary protection metric is the cosine similarity $CS(x, y)$ between the protected source x and the swapped output y on the QMagFace evaluator; a *lower* value means the swap fails to carry the protected identity. We report raw id-sim and its drop Δ relative to the unprotected swap, since id-sim is directly comparable to the identity-similarity numbers reported by prior work [39]. Because swapped faces share structural similarity with the source, we additionally report a Normalized Cosine Similarity that centers the unprotected swapped-population mean m at zero:

$$NCS(x, y) = \begin{cases} \frac{CS(x,y)-m}{1-m} & \text{if } CS(x, y) \geq m \\ \frac{CS(x,y)-m}{1+m} & \text{if } CS(x, y) < m \end{cases} \tag{12}$$

A *lower* NCS likewise indicates better protection.

ii) **Attack Success Rate (ASR)**: This metric quantifies the ratio of successful defenses. A defense is considered successful if the identity similarity between the protected source and the swapped output falls below a threshold of 0.36 (indicating identity mismatch), measured on the held-out QMagFace evaluator. Higher ASR is better.

iii) **Perceptual Fidelity (PSNR / SSIM / LPIPS / ℓ_∞)**: We measure the visual fidelity of the *protected* image against the clean source with PSNR, SSIM, and LPIPS [46], and additionally report the ℓ_∞ pixel budget. ℓ_∞ is informative here because it separates a *bounded additive perturbation* (small ℓ_∞) from a *full regeneration* (large ℓ_∞), even when the latter scores well on LPIPS/SSIM.

iv) **Structural Preservation (Landmark NME)**: To quantify whether a defense alters facial *geometry*, we report the five-point landmark Normalized Mean Error between the clean and protected images (mean landmark displacement divided by the inter-ocular distance, in %). A method that preserves facial structure yields a low NME.

4.2.2. On the Reproducibility of GAN-based Defenses

Our primary comparison is against NullSwap [39], the state-of-the-art generator-based defense. Since no official implementation is released, we faithfully re-implemented it following the paper: the same generator/discriminator design, identity extractors (ArcFace + FaceNet), loss weights (λ_{id}=0.08, λ_{MSE}=1.8, λ_{LPIPS}=1.2, λ_D=0.1), and the Dynamic Loss Weighting schedule (α=3, β:0.5→2, k=30). Despite this, our best re-implementation attains only PSNR 29.2dB / SSIM 0.914 / LPIPS 0.079 on CelebA-HQ, far short of the reported 41.3dB / 0.986 / 0.005 (Table 1), and the resulting images exhibit clearly perceptible *structural* distortion of facial features rather than an imperceptible cloak. Constrained to a single RTX 3090 we train for 40 epochs at batch size 64, versus the paper's 60 epochs at batch size 256 on 8×A100; even so, the 12dB fidelity gap and the instability we observe are consistent with the well-documented fragility of adversarial GAN training and underline a reproducibility concern for such defenses. We therefore report our NullSwap re-implementation as an honest lower bound and, where possible, also compare against the numbers *originally reported* in [39].

4.2.3. Perceptual Fidelity and Structural Preservation

Table 1 compares the fidelity of the protected images. On LPIPS/SSIM our re-implemented NullSwap looks smoother than our additive perturbation, but this is misleading: its ℓ_∞ budget is 108/255 (42% of the dynamic range), confirming it *regenerates* the image rather than cloaking it, and its landmark NME (4.30%) is the largest among functional defenses—i.e., it moves facial geometry. Our method is a strictly bounded perturbation ($\ell_\infty \approx 26/255$, i.e. ϵ=8/255 inflated only by JPEG storage): its SSIM-regularized variant attains the highest PSNR (32.7dB) among methods that actually disrupt swapping, while its LPIPS-regularized variant reaches the lowest perceptual distance and the lowest structural distortion (LPIPS 0.133, NME 2.43%)—the regularizer simply trades one fidelity axis for another, and both preserve facial geometry far better than NullSwap. Anti-Forgery [37] is the most imperceptible in our hands (PSNR 48.3dB) but, as shown next, provides essentially *no* protection against modern face swappers, illustrating that fidelity alone is meaningless without protection.

4.2.4. Protection against Black-box Face Swappers

We now measure protection: a source is protected, swapped onto ten different targets with each of three black-box swappers (FaceDancer, SimSwap, HifiFace), and the swapped output is compared to the clean source with the held-out QMagFace evaluator. Lower identity similarity (id-sim) and higher ASR indicate stronger protection; Δ is the drop in id-sim relative to the unprotected swap. Table 2 reports the mean over the three swappers.

Our method provides clear protection (id-sim 0.270, Δ 0.128, ASR 0.76 on QMagFace), reducing identity leakage well below the unprotected baseline, whereas Anti-

Method	PSNR ↑	SSIM ↑	LPIPS ↓	ℓ_∞ ↓	NME(%) ↓
Ours (SSIM reg.)	**32.68**	0.899	0.166	27	3.59
Ours (LPIPS reg.)	31.41	0.827	0.133	**26**	**2.43**
NullSwap (our re-impl.)	29.20	0.914	0.079	108	4.30
NullSwap (reported [39])	41.31	0.986	0.005	–	–
Anti-Forgery (our re-impl.)	48.30	0.996	0.008	9	0.41

Table 1. Perceptual fidelity of the *protected* source image vs. the clean source on CelebA-HQ (992 samples). ℓ_∞ is the max per-pixel change in $[0, 255]$; large ℓ_∞ with low LPIPS indicates full regeneration rather than a bounded cloak. Landmark NME measures facial-geometry displacement (lower = better structural preservation). NullSwap's *reported* row is far above our faithful re-implementation, reflecting a reproducibility gap.

Method	id-sim ↓	Δ ↑	ASR ↑
No protection (baseline)	0.398	0.000	0.39
Anti-Forgery (our re-impl.)	0.398	0.000	0.39
Contrastive Attack (Ours)	0.270	0.128	0.76
NullSwap (our re-impl.)	**0.117**	**0.281**	**0.95**
NullSwap (reported [39])†	0.332	–	–

Table 2. Protection on CelebA-HQ, QMagFace evaluator, mean over FaceDancer/SimSwap/HifiFace × 10 targets. id-sim is the cosine similarity between the protected source and the swapped output; Δ is the drop vs. the unprotected swap; ASR uses threshold 0.36. †Reported NullSwap uses a different evaluator/swapper set (ArcFace/VGGFace; SimSwap/InfoSwap/UniFace/E4S/DiffSwap) and is shown only as an external reference.

Forgery yields $\Delta \approx 0$—its perturbation is stripped by the swappers' internal re-alignment, so despite the best fidelity it offers no real protection. Our re-implemented NullSwap attacks harder (Δ 0.28, ASR 0.95), and the gap widens on FaceNet (where NullSwap drives id-sim negative); we report this honestly. Two points reframe it, however. First, our protection is achieved with a bounded, reproducible perturbation, whereas NullSwap's strength comes from regeneration that distorts the face (Table 1, Fig. 2). Second, against the numbers *originally reported* for NullSwap (id-sim 0.332 [39]), our id-sim of 0.270 is competitive—even lower—although this crosses evaluator/swapper setups and should be read as indicative rather than a controlled comparison.

4.2.5. Cross-dataset (Out-of-distribution) Transfer

Since NullSwap is trained on CelebA-HQ, we test generalization on LFW [16] without any retraining or re-optimization of hyperparameters. LFW swaps already have low baseline identity similarity on QMagFace, so we read protection through Δ (the reduction beyond this baseline) rather than absolute ASR. Both methods weaken out of distribution: our attack retains a positive protection margin (Δ 0.049 on QMagFace, Δ 0.11 on FaceNet), while NullSwap remains stronger (Δ 0.11 / Δ 0.43). The point is not that we match NullSwap here, but that a training-free optimization over frozen surrogates transfers to an unseen dataset at all—whereas the learned generator's absolute margins also shrink markedly from their in-distribution values, consistent with dataset-specific overfitting.

4.2.6. Qualitative Analysis

Figure 3 shows the per-model cosine-similarity trajectory of our Contrastive Attack across the surrogate ensemble, which decreases consistently towards the antipodal region. Figure 4 isolates the role of our contrastive negatives: under plain gradient descent, a naive attack that targets the antipodal vector directly stalls at the gradient-vanishing saddle of Eq. 9 and leaves the source identity intact, whereas our contrastive objective escapes it and collapses the similarity.

Figure 2 contrasts the protected images produced by our method and by our NullSwap re-implementation on the samples where NullSwap distorts geometry most. Although our additive perturbation is more visible as high-frequency noise, it leaves the facial structure intact—eyes, nose, mouth, and jawline remain aligned with the clean source. NullSwap, being a full regeneration, instead re-synthesizes the face and visibly alters identity-bearing structure (nose shape, mouth, and overall proportions), which explains its high ℓ_∞ and landmark NME despite competitive LPIPS/SSIM. This is precisely the failure mode that makes GAN-based cloaks unreliable for faithful identity protection. Figures 5 and 6 present the downstream swap results on the out-of-distribution LFW set: swapping the protected source shifts the swapped identity away from the source relative to the clean-source swap, and this holds across FaceDancer, SimSwap, and HifiFace even though none is seen during optimization.

5. Conclusion

In this paper, we presented a novel **Contrastive Attack** framework designed to protect user identity against unauthorized face swapping without relying on specific victim images. By identifying the limitations of existing gradient-based attacks—specifically the "orthogonality curse" in high-dimensional latent spaces—we proposed a theoretically grounded strategy that sets the antipodal point of the source identity as the attack objective. To overcome the gradient vanishing problem inherent in this approach, we introduced a contrastive learning objective combined with facial attribute augmentation, ensuring robust optimization.

Our experiments show that the proposed method delivers competitive black-box protection while offering two properties that generator-based defenses lack: reproducibility and structural preservation. We faithfully re-implemented the state-of-the-art NullSwap and could not reproduce its reported fidelity (a 12dB PSNR gap), with the resulting model instead producing perceptible structural distor-

Figure 2. Protected images (no swap applied). **Left:** clean source. **Middle:** ours—a bounded additive perturbation that leaves facial geometry intact. **Right:** our NullSwap re-implementation—a full regeneration that visibly alters identity-bearing structure (nose, mouth, proportions), despite competitive LPIPS/SSIM. Rows are the samples with the largest NullSwap landmark displacement.

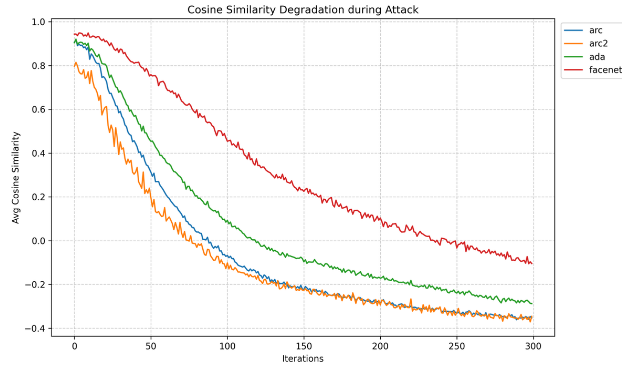

Figure 3. Cosine similarity trajectory during our **Contrastive Attack**. The similarity steadily decreases, indicating effective optimization towards the antipodal direction.

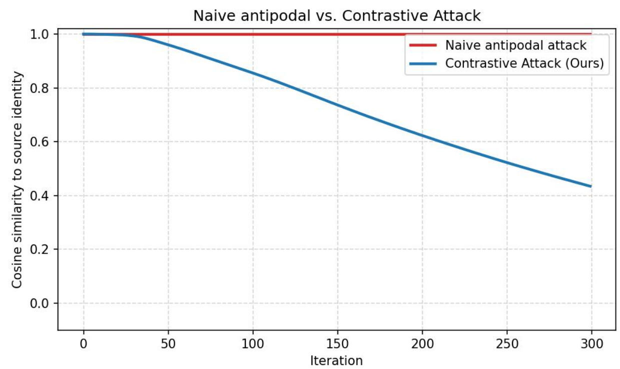

Figure 4. Naive antipodal attack vs. our **Contrastive Attack** under plain gradient descent (mean cosine similarity to the source identity over the surrogate ensemble, $\epsilon = 8/255$). Directly optimizing toward the antipodal vector $-\theta(x)$ stalls at the saddle predicted by Eq. 9, where the gradient magnitude $\propto \sin\theta \to 0$ at initialization, so the source identity is retained (flat red curve). Our contrastive negatives sit at a non-zero angle and supply an escape gradient, collapsing the similarity (blue). The sign-based MI-FGSM update used in our final attack mitigates this saddle for the naive case as well; the contrastive objective additionally removes any need for an external victim identity.

tions—an outcome consistent with the instability of adversarial GAN training. In contrast, our gradient-based attack is training-free and deterministic, applies a strictly bounded perturbation ($\ell_\infty = \epsilon$), and preserves facial geometry (lowest landmark NME), so the protected face remains semantically faithful to the original identity.

Although our approach introduces more visible high-frequency noise than an idealized generative cloak, this visibility is a bounded and predictable trade-off, and—unlike a full regeneration—it does not silently alter identity-bearing structure such as nose shape or facial hair. We believe this work provides a reproducible, victim-free baseline for adversarial defense and highlights the importance of differentiable preprocessing in attacking black-box face recognition models. Future work will focus on minimizing the perceptual visibility of these perturbations while maintaining protection.

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Figure 5. Qualitative protection on out-of-distribution **LFW** (FaceDancer). Columns: source, target, swap using the *clean* source, and swap using the *protected* source. Only the source differs between the last two columns, so any change in the swapped face is attributable to our protection: the protected-source swap shifts away from the source identity.

Figure 6. Protection generalizes across black-box swappers on **LFW**. For a single protected LFW source, the identity shift between the clean-source and protected-source swaps is visible for FaceDancer, SimSwap, and HifiFace, despite none of them being seen during optimization.

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